

How action potentials carry information

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The four worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 15 TO 18 OF WEEK 4

22 SLIDES · 4 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

An action potential is the same size every time a neuron fires one. That raises three questions, and they are the plan for this lecture. How can a signal that never changes size tell the brain that one touch is light and another is hard? How does it travel a meter along an axon without fading? And what decides how fast it gets there? We answer the first with the refractory periods and frequency coding, the second with local current, and the third with axon diameter and myelin. Then we change the conditions and predict what happens.

Here is the problem we solve by the end. A dentist injects lidocaine beside the nerve that serves the lower teeth. Within a few minutes the lip and the teeth on that side are numb, and the drill causes no pain. The patient can still feel the dentist pushing on the tooth. A few hours later all feeling is back. The drug never reached the pain receptors in the tooth, and they still respond to the drill.

You need three facts to make a first prediction. Lidocaine blocks voltage gated Na^+ channels. It stays close to where it was injected, partway between the tooth and the brainstem. And the endings of pain receptors respond to the drill with channels that are not voltage gated Na^+ channels. So graded potentials still form in the tooth, and action potentials still start in the axon near the tooth. They stop in the stretch of nerve beside the injection, and nothing reaches the brainstem. Pain is felt only when action potentials reach the brain, and one blocked stretch is enough to stop them. The harder question is why pressure still gets through, when the axons that carry it lie in the same nerve and the same drug. Write down a guess before you read on.

See it move: race two axons

SLIDE 15 OF THE DECK

This is a working model of two axons that are stimulated at the same moment. The clock shows real milliseconds. You can choose which two axons to compare, change the diameter of axon A and take its myelin off, change the middle third of both axons, and change how often stimuli arrive at the trigger zone. The strip at the bottom shows which stimuli fire an action potential and which ones land inside the absolute refractory period. Come back to it as you read the sections below.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-ap-conduction-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-ap-conduction-notes.html)

THREE THINGS TO TRY

- 1 Choose the motor fiber and the slow pain fiber and watch the clock. Axon A arrives in 11 ms and axon B in 1 s. Then take the myelin off axon A. It is still 15 times wider than axon B, and it now conducts at only 3.9 m/s. Compare that with worked problem 2.
- 2 Put the myelin back on axon A and press Remove the myelin, which strips only the middle third. Read the new arrival time. Then press Remove the myelin and warm the body, and last press Apply local anesthetic. Say in your own words why the action potential stops in each case.
- 3 Slide the stimuli up from 100 to 500 per second, and then to 800. Read how many action potentials are fired each time, and compare the ceiling with worked problem 1. Then shorten the absolute refractory period to 1 ms and read the ceiling again.

The refractory periods

SLIDES 3 TO 6 OF THE DECK

Start with what you already have from the last lecture. The voltage gated Na^+ channel has two gates, an activation gate and an inactivation gate, and so it has three states. When it is closed and ready, the activation gate is closed and the inactivation gate is open, and a depolarization to threshold will open it. When it is open, both gates are open, for less than a millisecond. Then the inactivation gate closes and the channel is inactivated. An inactivated channel cannot open, however strongly the membrane is depolarized. It resets to closed and ready only after the membrane has repolarized. Everything in this lecture follows from that one fact.

A refractory period is a stretch of time after an action potential during which that patch of membrane is harder, or impossible, to fire again. The absolute refractory period starts when the membrane reaches threshold and lasts about 1 to 2 ms, which is most of the action potential. During it the Na^+ channels are open at first and then inactivated. No stimulus of any size can start a

second action potential, because a stimulus works by opening closed channels, and there are none.

The relative refractory period follows and lasts a few more milliseconds. Some Na^+ channels have reset and some have not. The voltage gated K^+ channels are still open, so K^+ is still leaving, and the membrane is hyperpolarized, farther from threshold than usual. Two things now work against a stimulus: open K^+ channels let positive charge leave as fast as the stimulus brings it in, and the membrane has farther to go. A second action potential is possible, but it takes a larger than normal stimulus, and it may be smaller than normal because only the channels that have reset can take part. An everyday comparison is a toilet that has just been flushed. While the tank is empty, pressing the handle does nothing, however hard you press. While it is refilling you can flush again, but the flush is weaker.

Refractory periods do three jobs. They keep action potentials separate, so they never overlap or add together the way graded potentials do. They set a maximum firing frequency. And they make conduction one way, because the membrane just behind a traveling action potential is refractory.

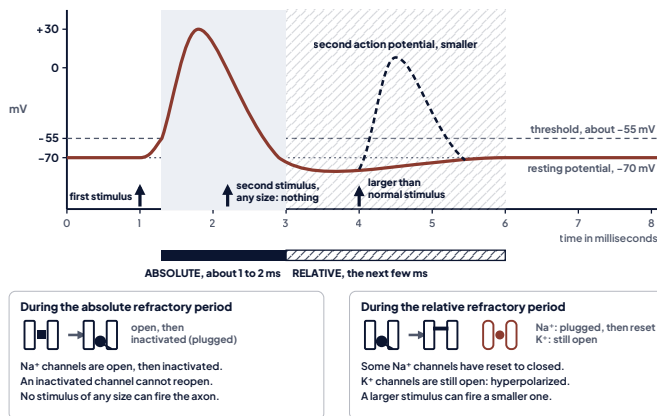


Figure 1. The two refractory periods, with the state of the channels during each one and the result of a second stimulus.

WORKED PROBLEM 1

An axon has an absolute refractory period of 2 ms. What is the largest number of action potentials it can send in one second? What if the refractory period were 1 ms?

maximum frequency = $1 \div \text{absolute refractory period}$, with the period in seconds.

THE WORK

- Put the refractory period into seconds. 2 ms = **0.002 s**.
- One action potential needs 0.002 s before the next one can start, so the most that fit into one second is $1 \div 0.002 = \mathbf{500 \text{ action potentials per second}}$.
- Repeat it for an absolute refractory period of 1 ms. $1 \div 0.001 = \mathbf{1,000 \text{ per second}}$. A shorter refractory period gives a higher ceiling.
- Read the answer as a sentence: however strong the stimulus is, this axon cannot send more than 500 action potentials in one second, because for 2 ms after each one its sodium channels cannot open.
- This is a ceiling and not a typical rate. Most neurons fire far below it, commonly well under 100 per second, because the relative refractory period comes next and a stimulus has to be unusually strong to fire the axon during it.

Absolute refractory period

About 1 to 2 ms from threshold. Na^+ channels are open and then inactivated, so no stimulus of any size can fire a second action potential.

Relative refractory period

The next few milliseconds. Some Na^+ channels have reset, K^+ channels are still open and the membrane is hyperpolarized. A larger than normal stimulus can fire a smaller than normal action potential.

Inactivated

The state of the voltage gated Na^+ channel after it has opened. It cannot reopen until the membrane repolarizes and the channel resets.

Maximum firing frequency

$1 \div \text{the absolute refractory period}$. With 2 ms it is 500 per second. Most neurons fire far below their ceiling.

How a neuron codes the strength of a stimulus

SLIDES 7 TO 8 OF THE DECK

Because every action potential is the same size, strength cannot be coded in the size of the signal. It is coded in the number of action potentials per second. A stronger stimulus opens more channels in the receptor ending, so it produces a larger graded potential, and one that lasts longer. At the trigger zone, a graded potential that is still above threshold when the absolute refractory period ends fires another action potential, and the larger it is, the sooner in the relative refractory period it can do that. So a larger, longer graded potential produces more action potentials per second. This is frequency coding. The ceiling is the one set by the

absolute refractory period, and the length of the burst codes how long the stimulus lasted.

At the axon terminal, each action potential releases about the same amount of neurotransmitter, so more action potentials per second release more neurotransmitter. That is how the next cell receives the information about strength. The next lecture covers how this release works.

A single axon has only that one way to report a stronger stimulus. The nervous system as a whole has a second way. A stronger stimulus reaches more receptors, including ones with a higher threshold, so more neurons fire. This is called recruitment, or population coding. The brain reads both: how fast each axon is firing, and how many axons are firing. Recruitment matters most at high intensities. An axon with a 2 ms absolute refractory period that is already firing at 400 per second cannot double its rate, because its ceiling is 500, but a stimulus twice as strong still recruits receptors that were silent before.

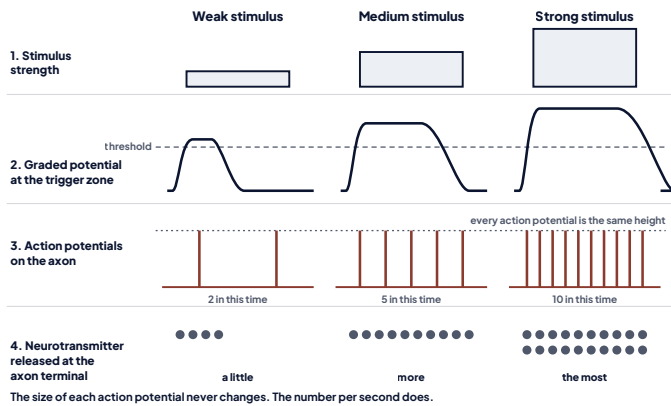


Figure 2. One sensory neuron and three strengths of stimulus. Read each column from top to bottom.

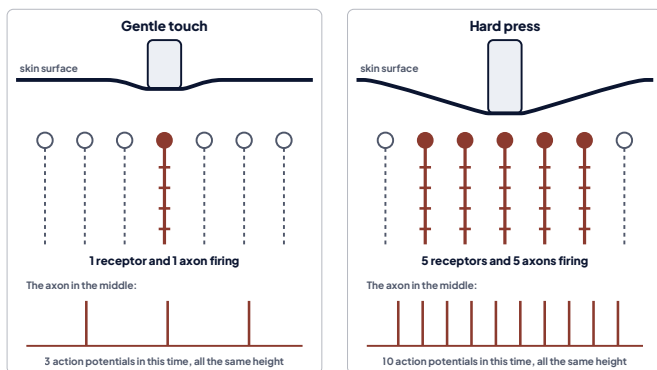


Figure 3. The same patch of skin touched gently and pressed hard.

Frequency coding

Signaling the strength of a stimulus by the number of action potentials per second, because their size is fixed.

Recruitment (population coding)

Signaling a stronger stimulus by the number of neurons that fire.

Trigger zone

The part of the neuron with the highest density of voltage gated Na^+ channels, where graded potentials are turned into action potentials.

How the action potential moves along the axon

SLIDES 9 TO 10 OF THE DECK

Stop a traveling action potential at one moment and look at the membrane. In the active zone, Na^+ has entered and the inside of the membrane is positive, while the neighboring stretches are still negative inside. Positive charge flows along the inside of the axon toward the negative regions on both sides. This flow of charge is called local current. Ahead of the action potential, the local current depolarizes resting membrane. When that patch reaches threshold, its own Na^+ channels open and it fires a full action potential, and then it does the same for the patch after it.

Behind the action potential, the local current arrives at membrane that fired a moment ago. Its Na^+ channels are inactivated and its K^+ channels are open, so it is refractory and cannot fire. By the time it has recovered, the active zone has moved too far away to reach it. That is why an action potential that starts at the trigger zone travels only toward the axon terminal. The axon itself has no preferred direction. If an axon is stimulated in the middle in the laboratory, two action potentials travel away from the electrode in opposite directions, because the membrane on both sides is at rest.

The action potential that reaches the axon terminal is therefore not the one that left the trigger zone. It is a copy, made fresh by each patch of membrane along the way, using the Na^+ and K^+ gradients at that patch. That is what regenerated means, and it is why the signal is the same size a meter away as it was at the start. A graded potential is different. It spreads by local current only, nothing renews it, charge leaks out through the membrane as it goes, and it is gone within a few millimeters. A row of falling dominoes is a fair comparison on two points: each domino falls with its own stored energy, so the last falls as hard as the first, and a domino that has fallen cannot fall again until it is set back up.

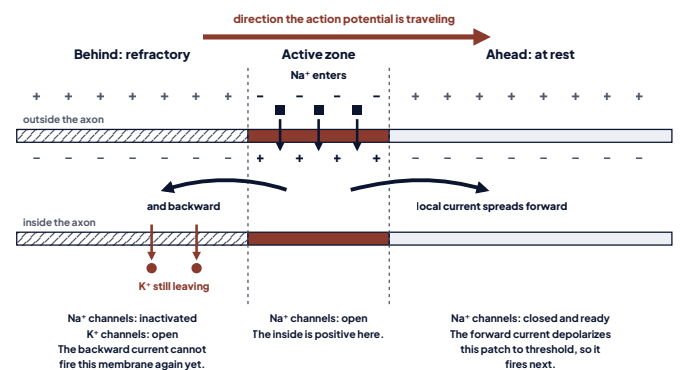


Figure 4. Local current spreads forward and backward. Only the forward direction finds membrane that can fire.

Local current

The flow of positive charge along the inside of the axon from the active zone toward neighboring membrane that is still negative inside.

Regenerated

Made again at each patch of membrane. The action potential does not decrease with distance because each patch fires its own.

Conduction

The movement of an action potential along an axon at full size.

What sets conduction velocity

SLIDES 11 TO 14 OF THE DECK

Two things decide how fast an axon conducts. The first is diameter. Local current flows through the cytoplasm, and a wider axon has more room for charge to flow, so its internal resistance is lower, in the same way that a wide pipe carries water more easily than a narrow one. The local current reaches farther ahead before it fades, so membrane farther down the axon is brought to threshold sooner. The second is how much current leaks out through the membrane on the way, because current that leaks out is lost to the job of depolarizing the next patch. Myelin, which is many layers of glial cell membrane wrapped around the axon, raises the resistance of the membrane so very little leaks out. It also means less charge is needed to change the voltage across that stretch. Both effects let local current spread farther and faster.

Myelin gives far more speed for the space it takes up. The squid giant axon has no myelin. It is about 500 μm across and conducts at about 25 m/s. Your largest myelinated axons are about 20 μm across and conduct at up to about 120 m/s. Area goes with the square of the diameter, and $(500 \div 20)^2$ is 625, so about 625 of your axons fit in the space of one squid axon, and each is nearly five times faster. That is what lets a nerve as thick as a finger carry many thousands of fast axons down a limb.

In an unmyelinated axon, voltage gated channels are spread along the whole membrane, and every patch has to depolarize and fire in turn. This is continuous conduction, and it is slow because opening channels takes time at every patch. In a myelinated axon the voltage gated Na^+ channels are concentrated at the nodes of Ranvier. Local current from one node flows inside the axon, under the myelin, and reaches the next node still strong enough to bring it to threshold, so the action potential is regenerated only at the nodes. This is saltatory conduction, from the Latin word for jumping. Nothing actually jumps. What travels between the nodes is local current, a very fast spread of charge along the inside of the axon. The time is spent at the nodes, and there are far fewer of those stops. It also costs less ATP, because Na^+ enters and K^+ leaves only at the nodes, so the Na^+/K^+ pump has far fewer ions to move back.

Real human nerve fibers conduct at anywhere from about 0.5 to about 120 m/s. Large myelinated A alpha fibers, 12 to 20 μm , conduct at 70 to 120 m/s and carry commands to skeletal muscle. A beta fibers, 5 to 12 μm , carry touch and pressure at 30 to 70 m/s. Thin myelinated A delta fibers, 2 to 5 μm , carry fast, sharp pain and cold at 12 to 30 m/s. Unmyelinated C fibers, 0.4 to 1.2 μm , carry slow, aching pain and warmth at 0.5 to 2 m/s. These are approximate ranges, and books differ a little on the boundaries. Two rough rules reproduce them: for a myelinated axon, velocity in m/s is about 6 times the diameter in μm , and for an unmyelinated axon it is about the square root of the diameter in μm . To rank any set of axons, put the myelinated ones first, and within each group put the larger diameter first.

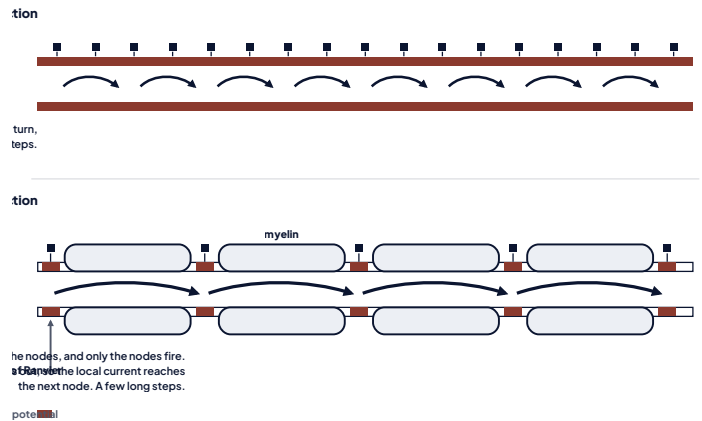


Figure 5. Two axons of the same diameter. The shaded membrane is the membrane that depolarizes and fires.

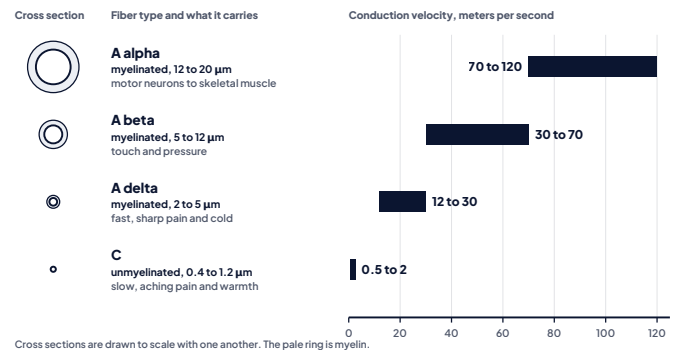


Figure 6. Approximate ranges for human nerve fibers.

WORKED PROBLEM 2

Sensory axons from your toe run about 1 m to the spinal cord. How long does an action potential take in a large myelinated fiber at 100 m/s, and in a small unmyelinated fiber at 1 m/s? Then explain why a stubbed toe hurts twice.

time = distance $\div$ velocity.

THE WORK

- The relationship is time = distance $\div$ velocity.
- Large myelinated fiber: $1 \text{ m} \div 100 \text{ m/s} = 0.010 \text{ s} = \mathbf{10 \text{ ms}}$.
- Small unmyelinated fiber: $1 \text{ m} \div 1 \text{ m/s} = \mathbf{1 \text{ s}}$. That is 100 times longer, for the same stretch of leg.
- Now the two kinds of pain fiber. A delta fibers are thin and myelinated. At about 20 m/s, 1 m takes $1 \div 20 = 0.05 \text{ s} = \mathbf{50 \text{ ms}}$. C fibers are unmyelinated. At about 1 m/s, 1 m takes $\mathbf{1 \text{ s}}$.
- So when you stub your toe, the action potentials in the A delta fibers reach the spinal cord about a second before the ones in the C fibers. You feel a sharp, well located pain first, and a dull, aching, spreading pain about a second later. These are called first pain and second pain. Both sets of action potentials started at the same moment.
- The withdrawal reflex uses the faster fibers, which is why your foot has already moved before the ache begins.

Continuous conduction

Conduction in an unmyelinated axon, where every patch of membrane fires in turn.

Saltatory conduction

Conduction in a myelinated axon, where the action potential is regenerated only at the nodes of Ranvier and local current carries the depolarization between them.

Node of Ranvier

A short bare gap between segments of myelin, where voltage gated Na^+ channels are concentrated.

First pain and second pain

The sharp pain carried by A delta fibers, followed about a second later by the aching pain carried by slower C fibers.

Changing the conditions: myelin, potassium and calcium

SLIDES 16 TO 18 OF THE DECK

Remove the myelin from part of an axon, as happens in the central nervous system in multiple sclerosis. Where myelin is lost, local current leaks out through the bare membrane, so less of it reaches the next node. That node takes longer to reach threshold, and conduction slows. If too little arrives, the node never reaches threshold and conduction fails. The bare stretch cannot make up for it, because it was under myelin and has very few voltage gated Na^+ channels of its own. Warming makes this worse. If body temperature rises from 37.0°C (98.6°F) to 37.5°C (99.5°F), ion channels gate faster, the Na^+ channels at each node inactivate sooner, and each action potential is briefer and sends less current on. A healthy node receives several times more current than it needs, so it is unaffected. A node beyond a demyelinated stretch is receiving barely enough, so it fails. Cooling brings it back, and no new damage has occurred.

Change the potassium in the blood. Plasma K^+ is the outside K^+ for every neuron and muscle cell. In the resting potential lecture you calculated that 7 mM moves the resting potential to about -64 mV and 2.5 mM moves it to about -76 mV . Threshold is set by the Na^+ channels and stays at about -55 mV , so what changes is the gap between rest and threshold. Worked problem 3 follows both cases through.

Change the calcium in the blood. Ca^{2+} ions in the extracellular fluid bind to the outside of the voltage gated Na^+ channel, and while they are bound the channel needs a larger depolarization to open. With low extracellular Ca^{2+} the channels open more easily, so threshold moves closer to the resting potential. Nerves begin to fire without being stimulated, and the muscles they supply contract. The result is tingling, cramps and sustained spasm, called tetany. In Chvostek's sign, tapping the facial nerve in front of the ear makes the face twitch. In Trousseau's sign, a blood pressure cuff kept inflated on the arm for about three minutes brings on a spasm of the hand and wrist. With high extracellular Ca^{2+} the reverse happens, and nerve and muscle are less excitable. Students often expect low calcium to weaken muscle. The Ca^{2+} that makes skeletal muscle contract is stored inside the muscle cell, and a low plasma Ca^{2+} hardly changes that store. It also explains why calcium given into a vein helps in hyperkalemia: it moves threshold away from the depolarized resting potential and restores the gap.

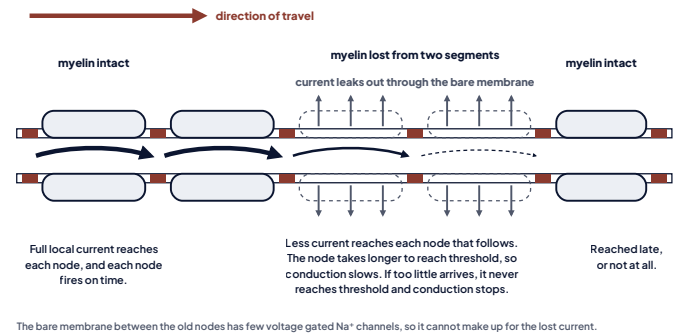


Figure 7. Demyelination. Less current reaches each node beyond the bare stretch.

Plasma potassium moves the resting potential. Threshold stays at about -55 mV .

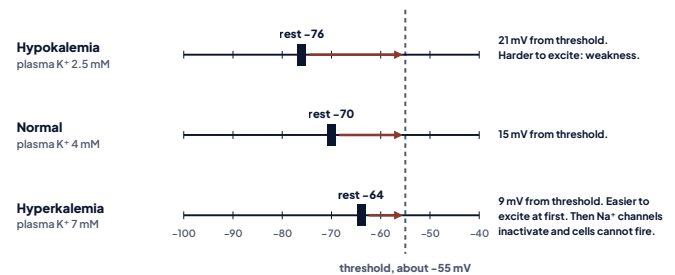


Figure 8. The gap between the resting potential and threshold at three plasma potassium values.

WORKED PROBLEM 3

For a plasma K^+ of 2.5 mM, 4 mM and 7 mM, work out how far the resting potential is from threshold and predict what happens to excitability. Explain both stages of hyperkalemia.

Resting potentials from the resting potential lecture: about -76, -70 and -64 mV. Threshold about -55 mV.

THE WORK

- 1 Normal, 4 mM: the resting potential is about -70 mV and threshold is about -55 mV, so the gap is **15 mV**. A graded potential has to depolarize the trigger zone by 15 mV to fire the neuron.
- 2 Hypokalemia, 2.5 mM: from the resting potential lecture, the resting potential is about -76 mV. The gap to threshold is $-55 - (-76) = 21 \text{ mV}$. The same graded potential that used to reach threshold now falls short, so neurons and skeletal muscle are **less excitable**. The patient is weak.
- 3 Hyperkalemia, 7 mM: the resting potential is about -64 mV. The gap is $-55 - (-64) = 9 \text{ mV}$. At first a smaller graded potential is enough, so cells are **more excitable**.
- 4 Now add what you know about the sodium channel. A membrane that stays depolarized holds some of its voltage gated Na^+ channels in the inactivated state, and an inactivated channel cannot open. The longer and the further the membrane is depolarized, the fewer channels are closed and ready.
- 5 With too few channels ready, the action potential rises slowly and is small, and if enough are inactivated the cell cannot fire at all. So hyperkalemia makes cells more excitable first and then less excitable, and what changes in between is the number of Na^+ channels that are available.
- 6 The Goldman equation gives -71 mV for the normal resting potential with the values from that lecture. It is rounded to -70 mV here. Books also vary by a few millivolts on threshold.

Hyperkalemia

High plasma K^+ . The resting potential depolarizes toward threshold. Cells are more excitable at first, and then Na^+ channels inactivate and cells cannot fire.

Hypokalemia

Low plasma K^+ . The resting potential hyperpolarizes, farther from threshold, and cells are less excitable.

Tetany

Sustained muscle spasm caused by nerves firing on their own, as in low extracellular Ca^{2+} .

Blocking the sodium channel, and the dentist's chair explained

SLIDES 19 TO 21 OF THE DECK

There are two different ways to interfere with Na^+ entry, and they give different results. Lower the Na^+ outside the axon, and E_{Na} falls, so the peak of the action potential is lower, but the channels still open and the action potential is still conducted. Block the channel, and there is no action potential at all. Local anesthetics such as lidocaine cross the membrane, enter the voltage gated Na^+ channel from the inside of the cell and block the pore. Tetrodotoxin blocks the same channel from the outside.

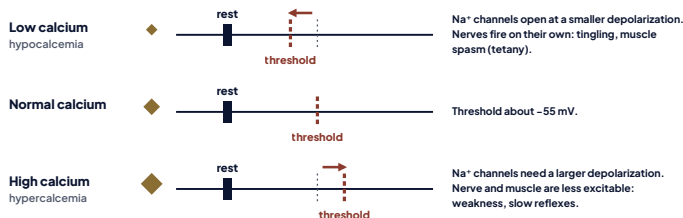
Now the dentist's chair. The pain receptors and the first stretch of axon work normally, so graded potentials still form and action potentials still start and travel toward the brainstem. In the injected stretch the Na^+ channels cannot open. Lidocaine binds most readily to channels that are open or inactivated, so axons that are firing are blocked sooner. Local current from the last working patch still flows into the blocked stretch, but no channels open, so it is only a graded depolarization and it fades within a few millimeters. Nothing reaches the brainstem, so there is no pain.

Small fibers are blocked before large ones. To stop a myelinated axon, the drug has to block about three nodes in a row. In thin A delta fibers the nodes are close together, so a short stretch of drug covers enough of them. In thick A beta fibers, which carry pressure, the nodes are up to a millimeter or two apart, and the same stretch of drug may not cover three. C fibers are thin and have no myelin to keep the drug out. That is why the patient feels pushing and no pain. This order is the usual teaching rule and it is a simplification, because where a fiber lies within the nerve and how often it is firing also matter. Everything returns when the drug diffuses away, because nothing was damaged.

Look at it as an experiment. The problem is that action potentials from the tooth reach the brain. The target variable is the number of Na^+ channels that can open in one stretch of axon. The intervention is a drug that blocks them. The mechanism is that the action potential cannot be regenerated in that stretch. The same mechanism predicts what happens if a large amount of the drug reaches the blood. Every neuron and every heart muscle cell has these channels, so the first effects are in the brain, with numbness around the mouth, ringing in the ears and then seizures, and in the heart conduction slows. This is why a local anesthetic is kept local.

Every perturbation in this lecture acted on the same thing: how many voltage gated Na^+ channels are closed and ready to open. Hyperkalemia inactivates them. Calcium changes how easily they open. A local anesthetic blocks them. Demyelination reduces the current that reaches them.

Extracellular calcium changes how easily voltage gated Na^+ channels open



Drawn to show direction only. The resting potential barely changes, because the resting membrane lets almost no calcium through.

Figure 9. Extracellular calcium changes how easily the Na^+ channels open. Direction only.

Demyelination

Loss of myelin from part of an axon. Current leaks out between nodes, and conduction slows or fails.

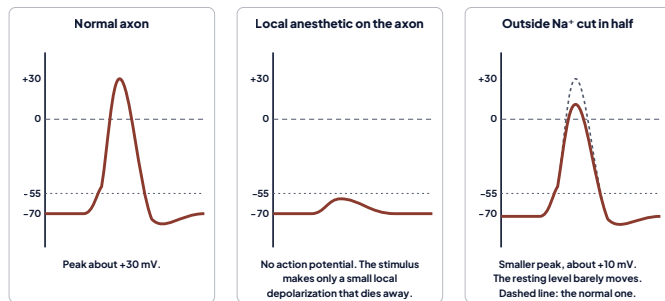


Figure 10. The same axon and the same stimulus under three conditions.

WORKED PROBLEM 4

An axon is bathed in fluid with 72.5 mM Na^+ , half the normal 145 mM. Inside stays at 15 mM. Calculate the new E_{Na} , predict what happens to the peak of the action potential, and compare that with an axon whose Na^+ channels are blocked.

$$E_{\text{ion}} = (61.5 \text{ mV} \div z) \times \log_{10}(\text{outside} \div \text{inside}), \text{ with } z = +1.$$

THE WORK

- 1 Normal sodium: $E_{\text{Na}} = 61.5 \times \log_{10}(145 \div 15) = 61.5 \times 0.985 = +61 \text{ mV}$.
- 2 Half the outside sodium: $E_{\text{Na}} = 61.5 \times \log_{10}(72.5 \div 15) = 61.5 \times 0.684 = +42 \text{ mV}$. Halving the outside concentration lowers E_{Na} by $61.5 \times \log_{10}(2) = 18.5 \text{ mV}$.
- 3 At the peak of an action potential the membrane is far more permeable to Na^+ than to anything else, so the peak follows E_{Na} . When E_{Na} drops by about 19 mV, the peak drops by about the same amount, from about +30 mV to about +10 mV. Hodgkin and Katz did this experiment on the squid giant axon in 1949, and the peak fell by about as much as the Nernst equation predicts.
- 4 The resting potential changes only a little, because the resting membrane is barely permeable to Na^+ . You saw the same thing in the resting potential lecture.
- 5 The action potential is smaller, but it is still there. The channels still open, Na^+ still enters, and each patch of membrane still fires the next. With a local anesthetic or tetrodotoxin the channels cannot open at all, so there is no action potential to conduct.

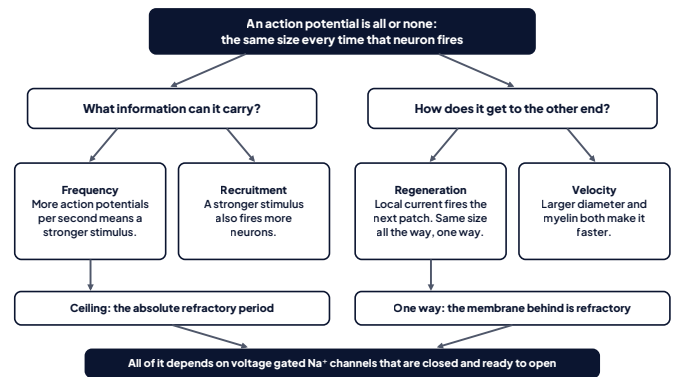


Figure 11. The whole lecture on one map. Read it from the top down.

Local anesthetic

A drug such as lidocaine that blocks voltage gated Na^+ channels from the inside of the cell, so no action potential is conducted past the blocked stretch.

Tetrodotoxin

A toxin that blocks the voltage gated Na^+ channel from the outside.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

What an ion is, and the charges on Na^+ , K^+ and Ca^{2+}
 Opposite charges attract, and current is a flow of charge
 Resistance: the harder it is for charge to flow, the higher the resistance

Chemistry review page

FROM EARLIER LECTURES

The resting membrane potential, the Nernst equation, and how plasma K^+ moves the resting potential (this week)
 Neurons and glia: axon, trigger zone, axon terminal, myelin and nodes of Ranvier (this week)
 Graded potentials and the action potential: threshold, all or none, the two gates of the Na^+ channel, tetrodotoxin (this week)

The anatomy you need is from the neurons and glia lecture: axon, trigger zone, axon terminal, myelin and node of Ranvier.

MATH

Dividing by a decimal: $1 \div 0.002 = 500$
 Changing milliseconds to seconds: $2 \text{ ms} = 0.002 \text{ s}$
 $\text{time} = \text{distance} \div \text{velocity}$
 The log button, for worked problem 4

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Competencies 19 and 20, the action potential simulation and the

nerve conduction measurement, are done in lab. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

15

Coding of stimulus intensity

Explain how the nervous system encodes stimulus strength when action potential amplitude is fixed.

Do one of the two. Head your paper with the number.

15-1

Draw three stimulus bars of increasing strength across the bottom of your box, weak, medium, and strong. Above each, draw the spike train one sensory neuron sends, as vertical marks on a time line. Keep every spike exactly the same height. One line: say precisely what changed between your three trains, and what could not have changed.

15-2

Draw a patch of skin with a stimulus pressed into it, gentle on the left and hard on the right. In each, draw how many receptors and how many axons get recruited, and draw the spike train on one of those axons underneath. One line: the nervous system has two ways to say louder, so name both from your drawing and say which one a single axon can do on its own.

16

Refractory periods

Distinguish the absolute from the relative refractory period by channel state and explain how the refractory period sets maximum firing frequency and prevents backward conduction.

Do one of the two. Head your paper with the number.

16-1

Draw an action potential tracing and bracket the absolute and relative refractory periods underneath it. Along each bracket, draw the state of the voltage gated sodium channels, inactivated and then recovering, and show whether the potassium channels are still open. Then draw a second stimulus arriving inside each period and show the result. One line: say why no stimulus at all, however large, works during the absolute period.

16-2

Draw an axon with an action potential halfway along it and an arrow for the direction it is traveling. Shade the membrane behind it and mark what state its sodium channels are in. Draw the local current spreading both forward and backward, and show why only one direction succeeds. One line: an absolute refractory period of about 1 to 2 ms sets a ceiling on firing rate, so work out that ceiling in spikes per second.

17

Conduction velocity

Rank axons by conduction velocity using diameter and myelination and contrast continuous with saltatory conduction.

Do one of the two. Head your paper with the number.

17-1

Draw four axons stacked vertically: large myelinated, small myelinated, large unmyelinated, and small unmyelinated. Rank them fastest to slowest and write a rough velocity beside each, using the real range of about 0.5 up to about 120 m per second. One line: say which buys more speed per unit of space taken up, diameter or myelin, and why that matters for a nerve running down a limb.

17-2

Draw one myelinated and one unmyelinated axon of similar diameter side by side, the same length. On each, shade where the membrane actually depolarizes, continuously along one and node to node on the other. Draw the local current loops on both. One line: saltatory means jumping, but the action potential does not really jump, so say what is actually moving between the nodes.

18

Effects of altered extracellular ions

Predict the effect of hyperkalemia and hypokalemia on resting potential and excitability and explain why local anesthetics that block sodium channels abolish the action potential.

Do one of the two. Head your paper with the number.

18-1

Draw a voltage scale from minus 100 to plus 30 mV with the resting potential at minus 70 and threshold at about minus 55 marked on it. Using the Nernst logic for potassium, draw where the resting potential moves in hyperkalemia and in hypokalemia, and show the remaining gap to threshold in each. One line: hyperkalemia first makes the cell more excitable and then less, so say what happens to the sodium channels in between.

18-2

Draw a normal action potential tracing, then draw the same axon after a local anesthetic such as lidocaine has blocked its voltage gated sodium channels, and show what is left of the trace. Beside it, draw the axon in low extracellular sodium and show what changes about the peak, not about the resting level. One line: say why blocking the channel abolishes the signal while low extracellular sodium only shrinks it.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · How action potentials carry information. Velocities, diameters and times are approximate teaching values. All figures are original drawings for this course.